1 Never-trained games: 207 of 240 receiver episodes improve

● real partner wins ● shuffled partner wins ○ exact tie (excluded) one dot = one receiver episode

	Into Quill		Into Ember	
Absent from every stage of training				
Code-word lookup	 18/20	$p = 2.0\times10^{-4}$	 17/20	$p = 0.001$
Digit strings	 20/20	$p = 9.5\times10^{-7}$	 20/20	$p = 9.5\times10^{-7}$
Translate	 21/21	$p = 4.8\times10^{-7}$	 18/19	$p = 3.8\times10^{-5}$
Run code	 18/19	$p = 3.8\times10^{-5}$	 18/21	$p = 7.4\times10^{-4}$
Multiple choice	 28/40	$p = 0.008$	 29/40	$p = 0.003$
Trained games, evaluated on held-out episodes				
Words	 20/20	$p = 9.5\times10^{-7}$	 20/20	$p = 9.5\times10^{-7}$
Numbers	 20/20	$p = 9.5\times10^{-7}$	 20/20	$p = 9.5\times10^{-7}$
Two-word phrases	 20/20	$p = 9.5\times10^{-7}$	 20/20	$p = 9.5\times10^{-7}$
Sums	 40/40	$p = 9.1\times10^{-13}$	 40/40	$p = 9.1\times10^{-13}$
Name + city	 40/40	$p = 9.1\times10^{-13}$	 40/40	$p = 9.1\times10^{-13}$
Copy a passage	 22/22	$p = 2.4\times10^{-7}$	 18/18	$p = 3.8\times10^{-6}$
Swap passages	 40/40	$p = 9.1\times10^{-13}$	 40/40	$p = 9.1\times10^{-13}$
Join two halves	 40/40	$p = 9.1\times10^{-13}$	 40/40	$p = 9.1\times10^{-13}$
Continue a text	 27/27	$p = 7.5\times10^{-9}$	 13/13	$p = 1.2\times10^{-4}$
Match the original reply	 18/18	$p = 3.8\times10^{-6}$	 22/22	$p = 2.4\times10^{-7}$
Answer, then talk	 16/19 1 ties	$p = 0.002$	 16/20	$p = 0.006$
Answer a question	 22/26	$p = 2.7\times10^{-4}$	 11/14	$p = 0.029$
Join two facts	 28/40	$p = 0.008$	 26/40	$p = 0.040$
Paraphrase	 21/21	$p = 4.8\times10^{-7}$	 19/19	$p = 1.9\times10^{-6}$
Premise + hypothesis	 20/40	$p = 0.563$	 22/40	$p = 0.318$
Solve a word problem	 20/20	$p = 9.5\times10^{-7}$	 20/20	$p = 9.5\times10^{-7}$
Write code	 21/22	$p = 5.5\times10^{-6}$	 17/18	$p = 7.2\times10^{-5}$
Follow an instruction	 19/19	$p = 1.9\times10^{-6}$	 21/21	$p = 4.8\times10^{-7}$

Exact one-sided sign tests; exact ties excluded. Each direction is tested separately.
Rhyme play was also held out. The mixed co_play row cannot isolate it, so it is omitted.

nl21 final · red = real partner has lower payload cross-entropy than its shuffled control